Course Code: 23ME101T

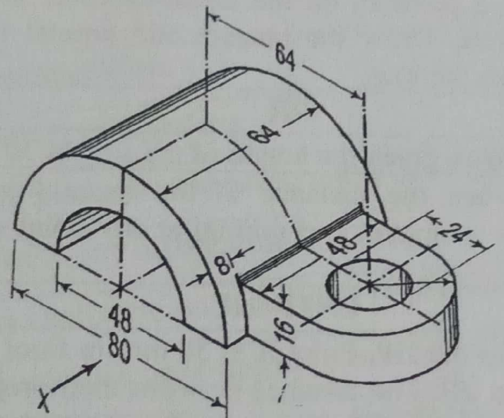
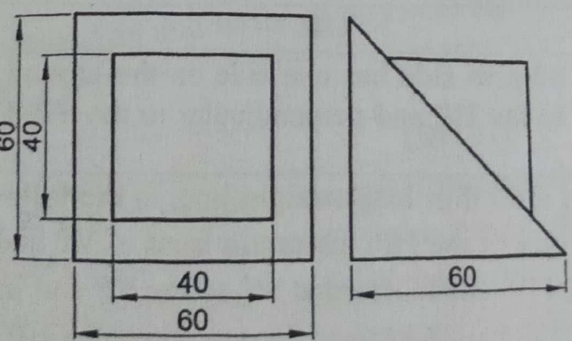
R23

PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)
I B.Tech I Semester Regular Examinations, January-2024
ENGINEERING GRAPHICS
(For CSE only)

Time: 3 hours **Max. Marks: 70**

Answer ONE Question from each Unit
All Questions Carry Equal Marks

Q. No.		Question	BTL	CO	Marks
UNIT – I					
1.		A Circle of 50 mm diameter rolls along a straight line without slipping. Draw the curve traced out by a point P, on the circumference, for one complete revolution of the circle. Draw the tangent and normal to the curve at a point on it 40mm from the line.	K3	CO1	14M
OR					
2.	a)	Construct a Regular Hexagon given the length of the side as 30 mm.	K3	CO1	6M
	b)	Construct a Parabola, when the distance of the focus from the directrix is 50 mm. Draw a tangent and normal at any point on the curve.	K3	CO1	8M
UNIT – II					
3,	a)	Two points A and B are in the HP. Point A is 30 mm in front of the VP, while B is behind the VP. The distance between their projectors is 75 mm and the line joining their top views makes an angle of 45^0 with xy. Find the distance of the point B from the VP.	K3	CO2	6M
	a)	A 90 mm long line is parallel to and 25 mm in front of the VP. It's one end is in the HP, while the other is 50 mm above the HP. Draw its Projections and find its inclination with HP.	K3	CO2	8M
OR					
4.	a)	A regular Pentagon of 30 mm side has one side on the ground. It's plane is inclined at 45^0 to the HP and perpendicular to the VP. Draw its Projections.	K3	CO2	6M
	b)	Draw the Projections of a 60 mm long straight line, in the following positions: (i) Perpendicular to the HP, 20 mm in front of VP and one end is 15 mm above the HP. (ii) Inclined at 60^0 to the VP and its one end is 15 mm in front of it: Parallel to and 25 mm above the HP.	K3	CO2	8M
UNIT – III					
5.		A square pyramid of base side 40mm and axis 55mm is resting on one of its triangular faces on the H.P. A vertical plane containing the axis is inclined at 45^0 to the V.P. Draw its projections.	K3	CO3	14M
OR					
6.		A pentagonal prism, side of the base 30 mm and axis 50 mm long, axis makes an angle of 60^0 to HP and the axis is parallel to VP. Draw its projections.	K3	CO3	14M

		UNIT – IV			
7.		A cylinder of 40 mm diameter, 60 mm height and having axis vertical, is cut by a sectional plane, perpendicular to the VP, inclined at 45^0 to HP and intersecting the axis 32 mm above the base. Draw its front view, sectional top view and true shape of the section.	K3	CO4	14M
		OR			
8.	a)	Draw the development of lateral surface of cylinder 30 mm and height 50 mm.	K3	CO4	6M
	b)	A pentagonal pyramid, edge of the base 25 mm and height 50 mm, rests on its base on HP with one of the base perpendicular to VP. A sectional plane parallel to HP cuts the pyramid at 20mm from HP. Draw the projections.	K3	CO4	8M
		UNIT – V			
9.		Draw the Front View, Top View, LHSV and RHSV for the given Isometric Projection in First Angle Projections. All Dimensions are in mm. 	K3	CO5	14M
		OR			
10.		Convert the given Orthographic Projection in First Angle Projection to Isometric Projection. All Dimensions are in mm. 	K3	CO5	14M